

The Light-independent Stage (Calvin Cycle)

Subject & topics: Biology | Biochemistry | Bioenergetics Stage & difficulty: A Level P3

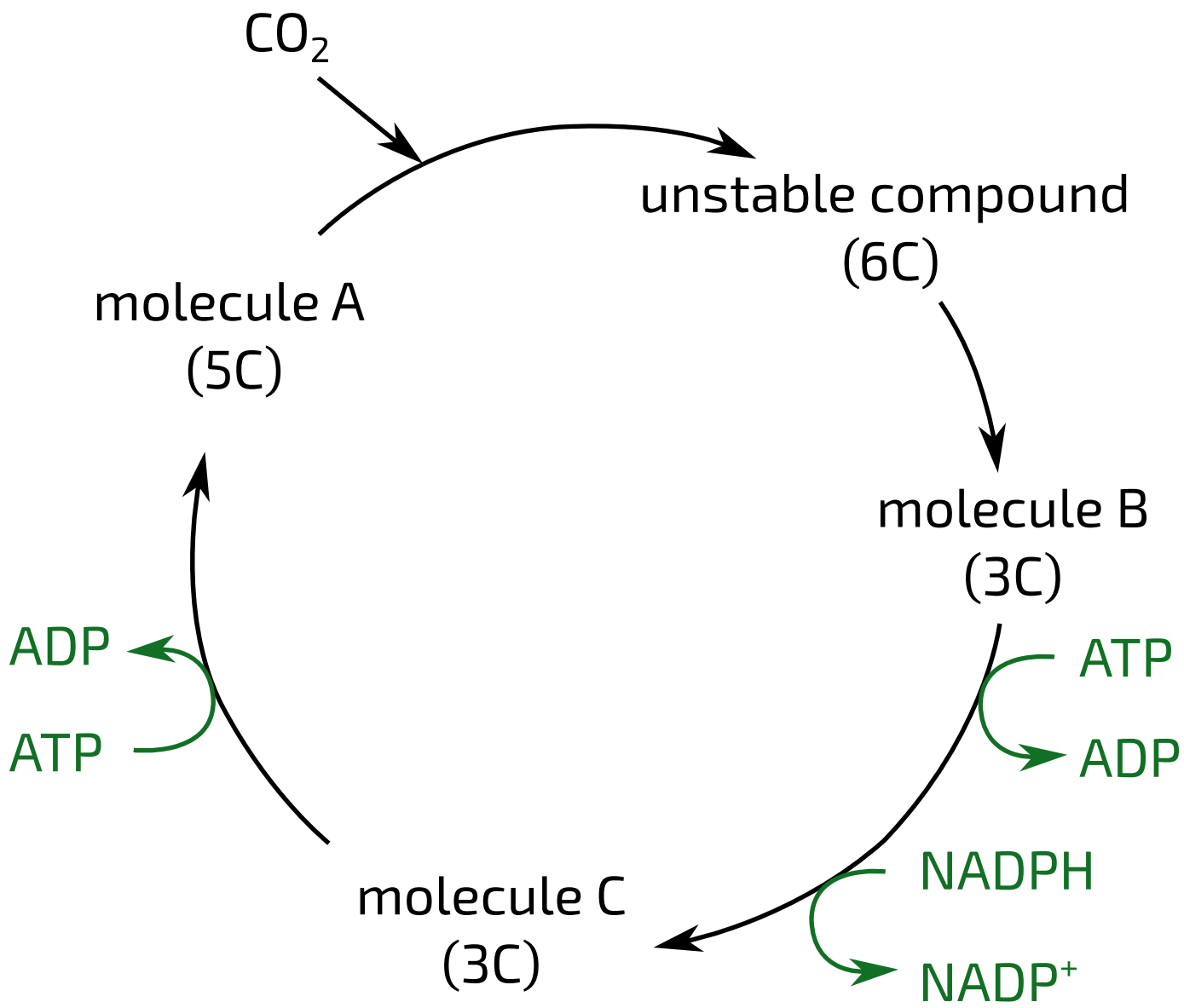


Figure 1: An overview of the light-independent stage of photosynthesis, also called the Calvin cycle. The number of carbons present in each molecule is shown e.g. 5C.

Part A Molecule A

What is the name of molecule A, the 5-carbon compound that reacts with carbon dioxide to form an unstable 6-carbon compound?

What is the name of the enzyme that catalyses this reaction?

Part B

Molecules B and C

What is the name of molecule B, the 3-carbon compound that is reduced to form molecule C?

What is the name of molecule C, the 3-carbon compound that is used in the formation of sugars, lipids, and amino acids - as well as in the regeneration of molecule A?

Part C

Numbers of molecules

For each molecule of CO_2 that is fixed (i.e. reacts with molecule A to form the unstable 6C compound), two copies of molecule C are produced.

If 5 out of every 6 copies of molecule C are used in the regeneration of molecule A, how many molecules of CO_2 are required to produce one molecule of glucose?

How many molecules of ATP would this require?

Note that one molecule of ATP is required to regenerate each copy of molecule A, and one molecule of ATP is required to convert molecule B into molecule C.

Adapted with permission from OCR A Level November 1999, Central Concepts in Biology, Question 5

Question deck:

STEM SMART Biology Week 35 - Photosynthesis 2

Krebs Cycle vs Calvin Cycle

Subject & topics: Biology | Biochemistry | Bioenergetics Stage & difficulty: A Level P3

Part A

Comparison table

Fill in the table below to compare the Krebs cycle to the Calvin cycle in eukaryotic cells.

	The Krebs cycle	The Calvin cycle
location		
electron carriers: reduced or oxidised		
carbon dioxide: used or produced		
ATP: used or produced		

Items:

- cytoplasm mitochondrial matrix mitochondrial inner membrane chloroplast stroma chloroplast thylakoid membrane
- reduced oxidised used produced

Part B

Electron carriers

What is the name of the main electron carrier in aerobic respiration?

What is the name of the main electron carrier in photosynthesis?

Part C

Limiting factors

Which of the following may act as limiting factors in the Krebs cycle? Select all that apply.

- ☐ CO₂ availability
- ☐ NAD⁺ availability
- ☐ NADH availability
- ☐ NADP⁺ availability
- ☐ NADPH availability
- ☐ ATP availability
- ☐ ADP availability

Which of the following may act as limiting factors in the Calvin cycle? Select all that apply.

- ☐ CO₂ availability
- ☐ NAD⁺ availability
- ☐ NADH availability
- ☐ NADP⁺ availability
- ☐ NADPH availability
- ☐ ATP availability
- ☐ ADP availability

Adapted with permission from OCR A Level January 2002 Central Concepts in Biology Question 2

Question deck:

STEM SMART Biology Week 35 - Photosynthesis 2

Leaf Discs Experiment

Subject & topics: Biology | Biochemistry | Bioenergetics

Stage & difficulty: A Level C1

Leaf discs were cut from destarched plants and placed either in water or in a glucose solution, and were exposed to air enriched with carbon dioxide.

The discs were kept under these conditions for 24 hours at 20 °C in either light or dark.

At the end of 24 hours the leaf discs were tested for starch. The results are shown in the table below.

	Discs in water	Discs in glucose solution
Light	✓	✓
Dark	✗	✓

Part A

Respiration and photosynthesis

ATP is required in the synthesis of starch. ATP is produced by the phosphorylation of ADP during both respiration and photosynthesis.

In the table below, show which process(es) was/were responsible were responsible for producing ATP in each experimental group. If neither process occurred, select "neither".

	Discs in water	Discs in glucose solution
Light		
Dark		

Items:

- photosynthesis
- respiration
- neither
- both respiration and photosynthesis

Part B

Testing for starch

What is the name of the test that would most likely be used to test the leaf discs for starch?

Part C

Amino acid synthesis

The triose phosphate produced in the light-independent stage of photosynthesis can be used to produce glucose and other carbohydrates, but it can also be used to produce amino acids.

In addition to those found in carbon dioxide and water, which element (that is present in all amino acids) do plants need to produce amino acids?

Adapted with permission from OCR A Level June 2001, Central Concepts in Biology, Question 5

Question deck:

STEM SMART Biology Week 35 - Photosynthesis 2

Chloroplasts and Mitochondria

Subject & topics: Biology | Biochemistry | Bioenergetics Stage & difficulty: A Level P3

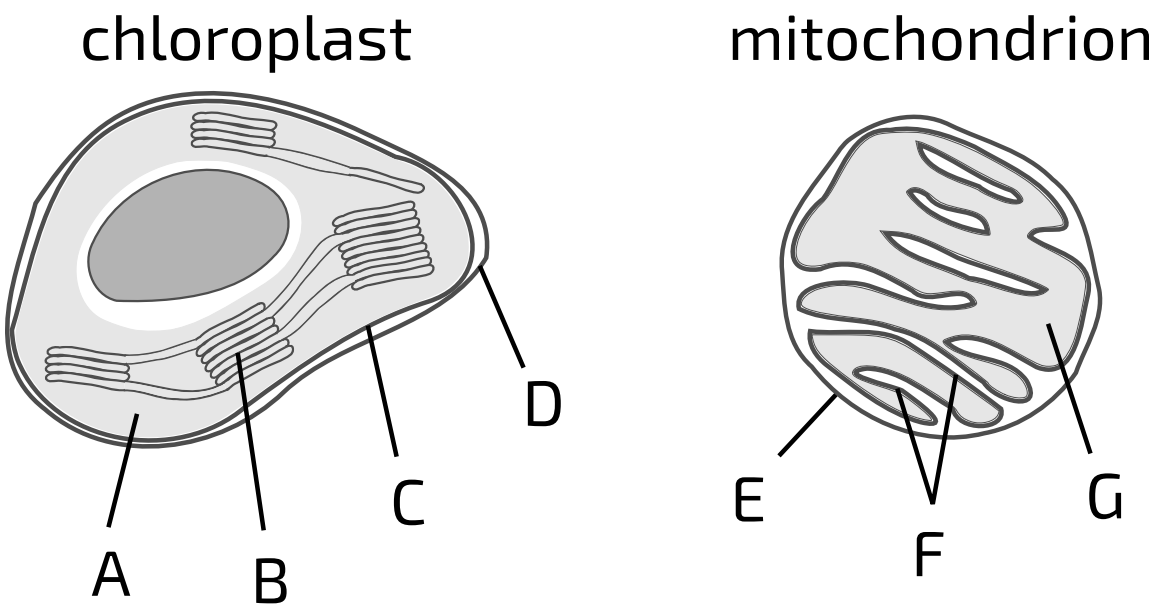


Figure 1: A diagram of a chloroplast and a mitochondrion. Different parts of the two organelles are labelled (A-G). The organelles are not shown to scale. The dark, oval-shaped structure in the chloroplast is a starch grain.

Part A

Label the diagram

Match the descriptions to the labels in the table below.

Label	Description
A	
B	
C	
D	
E	
F	
G	

Items:

- stroma

cristae (folds of the mitochondrial inner membrane)

chloroplast inner membrane

matrix

chloroplast outer membrane

thylakoid membrane

mitochondrial outer membrane

Part B

Stages of photosynthesis

Match the labels from **Figure 1** to the following processes, to show where in the organelle they occur (or select "none of the above" if they do not occur in either organelle).

- The light-dependent stage of photosynthesis:
- The light-independent stage of photosynthesis:

Items:

- A

B

C

D

E

F

G

none of the above

Stages of aerobic respiration

Match the labels from **Figure 1** to the following processes, to show where in the organelle they occur (or select "none of the above" if they do not occur in either organelle).

- Glycolysis:
- The link reaction:
- Krebs cycle:
- Oxidative phosphorylation:

Items:

A

B

C

D

E

F

G

none of the above

Photosynthesis vs Respiration

Subject & topics: Biology | Biochemistry | Bioenergetics Stage & difficulty: A Level P3

Part A Processes

In the table below, identify whether the processes occur during respiration or photosynthesis, and give the specific stage during which that process occurs.

Process	Respiration or Photosynthesis	Stage
A 6-carbon compound goes through a series of reactions to produce a 4-carbon compound. CO ₂ is produced.		
Water is split into oxygen, hydrogen ions (protons), and electrons.		
Oxygen reacts with electrons and hydrogen ions (protons) to produce water.		
CO ₂ reacts with a 5-carbon compound to produce a 6-carbon compound, which then breaks down into two 3-carbon compounds.		
Glucose is broken down into two pyruvate molecules. NADH and ATP are produced.		
Pyruvate is used to make acetyl CoA. NADH and CO ₂ are produced.		

Items:

- respiration photosynthesis glycolysis the link reaction Krebs cycle oxidative phosphorylation
- the light-dependent stage the light-independent stage

Part B

Locations

In the table below, identify whether the stages are part of respiration or photosynthesis, and give the specific location of that stage.

Stage	Respiration or Photosynthesis	Location
the link reaction		
Calvin cycle		
Krebs cycle		
non-cyclic photophosphorylation		
oxidative phosphorylation		
glycolysis		

Items:

- respiration
- photosynthesis
- cytoplasm
- mitochondrial matrix
- mitochondrial inner membrane
- chloroplast stroma
- chloroplast thylakoid membrane

Part C

ATP production

Which of the following stages directly result in the production of ATP? Select all that apply.

- ☐ glycolysis
- ☐ the link reaction
- ☐ Krebs cycle
- ☐ oxidative phosphorylation
- ☐ the light-dependent stage
- ☐ the light-independent stage

Question deck:

STEM SMART Biology Week 35 - Photosynthesis 2

Daylight Needed for Growth

Subject & topics: Biology | Biochemistry | Bioenergetics **Stage & difficulty:** A Level C3

Plants carry out both respiration and photosynthesis. In most plants, the rate of photosynthesis will be higher than the rate of respiration during the day, but lower during the night. In order for a plant to grow (produce biomass) over a period of time, the overall rate of photosynthesis must be greater than the overall rate of respiration during this period.

A particular plant has a total leaf surface area of 15 m^2 . During daylight, the rate of photon absorption by these leaves is $100 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$, and for every $1 \mu\text{mol}$ of photons absorbed, $0.04 \mu\text{mol CO}_2$ are fixed.

The plant respire at an average rate of 0.5 mol glucose per day (24 hours).

Calculate the minimum number of hours of daylight per day required for this plant to grow (produce biomass) each day.

Assume that respiration only occurs aerobically and that all CO_2 fixed is converted into glucose by the Calvin cycle.

Give your answer to the nearest hour.

Created for isaacscience.org by Lewis Thomson

Question deck:

STEM SMART Biology Week 35 - Photosynthesis 2

Photosynthesis Summary

Subject & topics: Biology | Biochemistry | Bioenergetics

Stage & difficulty: A Level P3

Part A

Reducing chlorophyll

Which process releases electrons that return chlorophyll molecules to their reduced state?

- ☐ activation of photosystem I
- ☐ active transport of hydrogen ions across the thylakoid membrane
- ☐ carbon fixation
- ☐ oxidation of NADPH (reduced NADP)
- ☐ phosphorylation of ADP
- ☐ photolysis of water
- ☐ reduction of glycerate phosphate to triose phosphate
- ☐ regeneration of ribulose biphosphate (RuBP)

Part B

Linking stages

Which of the following are products of the light-dependent stage that are used in the light-independent stage? Select all that apply.

- ☐ ADP
- ☐ ATP
- ☐ CO₂
- ☐ glucose
- ☐ NADP⁺
- ☐ NADPH (reduced NADP)
- ☐ O₂
- ☐ ribulose biphosphate (RuBP)

Part C

Radioactive labelling

Carbon dioxide labelled with ^{14}C has been used to identify the intermediate compounds in the light-independent stage of photosynthesis.

Which of the following compounds would be the first to contain the ^{14}C ?

- ☐ glycerate phosphate
- ☐ ribulose biphosphate (RuBP)
- ☐ glucose
- ☐ oxaloacetate
- ☐ triose phosphate (glyceraldehyde 3-phosphate)
- ☐ lactic acid
- ☐ acetyl CoA
- ☐ citrate

Question elements adapted with permission from CIE A Level November 2000 Paper 2 Question 26, CIE A Level June 1999 Paper 2 Question 26, and CIE A Level June 2001 Paper 2 Question 24

Question deck:

STEM SMART Biology Week 35 - Photosynthesis 2